

Systems of Representation and Omission: An Analytical Report on Dialogue Management, Socio-Technical Governance, and Algorithmic Arbitrage

The Semantics of Affirmation and Dialogue Management in Conversational AI

Human-computer interaction relies on Natural Language Processing (NLP) and Natural Language Understanding (NLU) to decode unstructured inputs and map them to deterministic intents. While explicit user commands containing fully qualified parameters are computationally straightforward, brief affirmative statements—such as "yes," "ok," "alright," or "exactly"—introduce significant ambiguity. Isolated confirmations possess high informational entropy; without an active context-tracking mechanism, the machine cannot determine what parameter, action, or proposal the user is validating.

To resolve this ambiguity, contemporary conversational architectures deploy advanced dialogue management systems. These systems track conversational states across multiple turns by assigning unique identifiers to specific interaction threads. For example, in automated home environments, a proactive system might trigger an inquiry when a user performs an action, such as turning on a light during the day. The query ("Do you want to turn off the kitchen light?") is bound to a persistent identifier, such as `conversation_id: question_ask`. When the user responds with a simple confirmation like "yes," the input processing engine routes the statement through an intent-matching script. By matching the user's brief response to the active `conversation_id`, the dialogue manager retroactively reconstructs the context, determining that the user is authorizing the specific light-deactivation action rather than attempting to initiate an unrelated or out-of-scope procedure.

A similar challenge occurs when users change context mid-conversation—such as shifting from an inquiry about local film showtimes to a query about the next day's weather. Under standard dialogue management protocols, if the system determines that the new query does not fit the active dialogue context, the platform's orchestration layer (such as the Google Assistant framework) automatically attempts to match the new intent and transition the user to a more appropriate conversational flow. If no match can be established, the session defaults back to the original action's context to prevent dialogue breakdown.

In conversational design, confirmation strategies are categorized based on risk, system confidence, and user friction.

Confirmation Type	Operational Mechanism	Representative Use Case	User Friction Level	Risk Mitigation Profile
Explicit Confirmation	Requires the user to actively confirm an understood parameter or	Completing financial transactions, deleting user data,	High	Maximum; crucial when the cost of system misunderstanding

Confirmation Type	Operational Mechanism	Representative Use Case	User Friction Level	Risk Mitigation Profile
	action before execution (e.g., "Did I get the message right?").	or confirming text messages to be sent on the user's behalf.		is severe.
Implicit Confirmation	Acknowledges the understood parameter within the next natural turn without pausing the dialogue for confirmation.	Confirming booking parameters (e.g., "booking for two people sitting together") and immediately asking the next question.	Low	Moderate; allows the user to speak up and correct errors without adding explicit conversational loops.
No Confirmation	Executes the requested action instantly with no verbal or textual acknowledgment.	Executing universal, high-confidence commands such as "stop," "cancel," or basic binary grammars.	None	Low; relies entirely on high-accuracy speech-to-text (STT) and NLU parsing.

To minimize friction and prevent dialogues from feeling overly robotic, designers apply specific conversational design principles. These principles are intended to match natural human cooperative behaviors, as described by philosopher Paul Grice's Cooperative Principle. First, systems must utilize plain, audience-centric language rather than reflecting the underlying database schemas. For instance, using "user account" instead of "user Id," or "where can I deliver this to" instead of "shipping address" improves naturalness. Second, conversational agents should not require users to memorize rigid command strings or "magic words". Instead of requiring a specific format like "Show me my sales for Q2 / 2021 as a pie with no_label double_y_axis," the system should understand natural phrasings like "Show me an overview of my sales for Q2" and render an appropriate visualization automatically. Third, dialogue interfaces must maintain high discoverability and usability by limiting choices to three options per turn, grouping additional parameters into a "more options" category, making the "Help" option mandatory, and avoiding confusing linguistic abbreviations. Advanced components also leverage tools like maxPrompts settings to automatically transition users to a help state if they repeatedly enter incorrect information.

Linguistic Pragmatics and Algorithmic Management in Organizational Contexts

The linguistic style of a conversational assistant must be tailored to the user's underlying intent and setting. Empirical research shows a clear division in user satisfaction when comparing abstract versus concrete language across hedonic and utilitarian settings.

Setting Type	User Operational Focus	Optimal Linguistic Style	Theoretical Grounding	Impact on User Evaluation
Hedonic Context	Emotional engagement, experiential value, and brand connection.	Abstract Language	Abstract framing better aligns with the open-ended, emotional needs of experiential consumption.	Higher overall satisfaction and stronger positive brand attitude.
Utilitarian Context	Task completion, information acquisition, and functional efficiency.	Concrete Language	Concrete phrasings provide clear, actionable details that directly support task execution.	Enhanced evaluation of utility and functional performance.

This functional alignment is equally critical when transitioning from consumer-facing assistants to internal organizational systems. In standard business settings, the introduction of Algorithmic Management (AM) tools—which automate scheduling, evaluation, and operational guidance—creates new dynamics in manager-algorithm interactions. Unlike platform-centered gig-economy firms where algorithms dictate tasks directly to workers, standard firms often experience friction between automated recommendations and human managerial judgment. Resolving this tension requires organizational designers to treat algorithms as collaborative entities rather than rigid rule engines, mirroring the turn-taking and context-aware capabilities found in advanced dialogue interfaces.

The Architecture of Omission: From Biological Platforms to Digital Observability

While the term "platform" typically refers to software infrastructures, a comparative analysis reveals that omission-based platforms operate across both biological and digital domains. In molecular biology, the concept of an "exon omission platform" describes a therapeutic gene-editing approach designed to bypass mutational hotspots in large genes. For example, in patients with Dystrophic Epidermolysis Bullosa (DEB), mutations in the COL7A1 gene (which has 118 exons) disrupt the production of anchoring fibrils. By delivering RNA-encoded gene-editing endonucleases via non-viral transfection methods, researchers can skip specific exons, such as the exon 73 splice acceptor, in primary cells like keratinocytes and fibroblasts. This biochemical omission bypasses the genetic mutation and restores functional protein creation.

Dimension	Exon Omission Platform (Biological)	Governance Omission Platform (Digital / Social)
Domain	Molecular Biology and Gene Therapy.	Digital Archiving, Sociology, and Political Science.
Core Target	Dysfunctional genetic sequences (e.g., COL7A1 exon 73 or 80 mutational hotspots).	Discrepancies between official administrative databases and actual lived experiences.
Mechanism	mRNA delivery of gene-editing endonucleases to skip mutated	Observability tools that track, measure, and expose

Dimension	Exon Omission Platform (Biological)	Governance Omission Platform (Digital / Social)
	sequences and restore protein function.	institutional gaps or erasures.
Socio-Political Focus	Addressing clinical pathologies like Dystrophic Epidermolysis Bullosa.	Resisting administrative erasure, identity reassignment, and systemic statelessness.

In the digital and socio-political domain, the concept of an omission platform is defined by the tension between recorded history and systemic erasure. In platformed societies, forgetting is rarely a passive decay of information; instead, it is often an active process driven by algorithmic omission, platform decay, or deliberate administrative erasure. This active erasure can obscure marginalized narratives and reshape what remains verifiable over time.

To measure and counter this erasure, visual communicator, graphic designer, and creative mentor Shane Bouel constructed the "OMISSION Platform" under the independent publication *Thoughtless Delineation*. Drawing on more than twenty years of professional experience in visual communication, multimedia design, and vocational education—including roles at CATC Design School and Keystone College—Bouel developed this platform as a governance observability tool.

The primary objective of this digital platform is to measure the gap between official administrative records and the lived realities of individuals who have suffered under state-managed classification schemes. Bouel's sociological and archival research focuses on systems that alter, hide, or reassign identity through paperwork without individual consent. This includes the historical legacies of systemic abuses like those associated with Georgia Tann, the case of Sasha Riley, and the thousands of international adoptees left in legal precarity.

A key focus of this work is the legacy of the Child Citizenship Act of 2000. Although passed to automate citizenship for foreign-born children adopted by U.S. citizens, the act omitted retroactive protections for those who had already reached adulthood by its effective date. This omission left thousands of adult adoptees without documented citizenship, exposing them to systemic legal threats, including deportation to countries where they had no linguistic or social ties.

These forms of institutional omission contrast sharply with active, community-led digital reclamation projects. For example, the Sámi visual repatriation initiatives colorize historical colonial photographs, transforming static, institutional archives into living, collaborative spaces of collective memory and cultural sovereignty. These distinct approaches highlight the ongoing tension between centralized archival control and grassroots digital reclamation.

The Legal Accountability of Platform Omission

When digital platforms fail to act, their omission can lead to direct civil liability. In legal frameworks like the Indonesian Civil Code (*Kitab Undang-Undang Hukum Perdata* or *KUH Perdata*), a platform's failure to moderate or remove harmful content after receiving user notifications is evaluated under tort law.

Specifically, Article 1365 of the *KUH Perdata* establishes that any unlawful act (*onrechtmatige daad*) that causes harm to another obligates the party at fault to compensate for the resulting damages.

Applying this statute to social media and content platforms requires assessing two main elements:

- **Fault (*Schuld*):** Evaluates whether the platform's omission to remove harmful or manipulated content—such as non-consensual deepfakes—constitutes intentional neglect (*opzet*) or actionable negligence (*culpa*) once a formal notification has been received.
- **Causality (*Causaliteit*):** Determines whether the psychological, financial, or reputational damage suffered by the victim is a direct result of the platform's delayed action, or if intervening factors broke the causal chain.

When platforms delay action for several weeks after receiving clear notifications, the prolonged visibility and redistribution of harmful material can exacerbate the victim's damages. This delay can elevate a simple failure to moderate into an actionable omission under Article 1365.

This liability framework emphasizes that platforms are not merely passive hosts; they are active operational entities with a duty of care to prevent their systems from causing harm.

Multi-Systemic Nexus Governance and Wicked Problems

Managing complex, interconnected resources—such as water, soil, energy, and waste—is increasingly recognized as a "wicked problem" characterized by conflicting goals, high systemic complexity, and informational uncertainty. Under the Nexus Approach, championed by international research institutes like UNU-FLORES, environmental resources are treated as deeply interconnected systems rather than isolated sectors. This approach aims to prevent solutions in one domain from creating unintended negative consequences in another.

A clear example of these system dynamics is found in China's Loess Plateau. In this region, a policy shifting steep cropland to forest cover succeeded in reducing soil erosion but also decreased the land available for agriculture. To offset this loss of arable land, local landholders increased their use of chemical fertilizers and pesticides on their remaining plots. Under conditions of water stress, this intensive farming degraded the local soil and groundwater, illustrating how an isolated environmental policy can trigger unexpected negative feedback loops across the broader water-soil-food nexus.

A similar challenge exists in urban environments, such as Accra and Kumasi in Ghana. In these rapidly growing cities, municipal solid waste and sanitation planning are often treated as narrow, technical infrastructure projects. This narrow focus frequently overlooks the social and behavioral factors needed for community-wide sanitation campaigns, reducing integrated nexus frameworks to mere tools for securing external funding rather than vehicles for sustainable, long-term resource recovery.

To address these informational gaps and ensure accountability in resource governance, researchers utilize multi-scalar monitoring platforms like the Nexus Observatory. In digital data environments, these verification efforts are supported by automated validation layers, such as the Nexus Truth Engine.

The Nexus Truth Engine does not act as an absolute authority on truth; instead, it serves as an automated validation framework that evaluates incoming data streams, sensor feeds, and AI-generated outputs against primary source records. This structured review helps minimize verification errors and ensures that public-facing reports remain grounded in verifiable facts.

Macro-Economic and Algorithmic Arbitrage in Indonesia (June 2026)

The real-world impacts of algorithmic recommendation engines, digital platforms, and macro-economic policies are clearly visible in the socio-economic landscape of Indonesia in mid-2026. During this period, the intersection of automated recommendation loops and national financial policy created a highly dynamic digital and economic environment.

In June 2026, international sports figures like David Alaba and Edin Dzeko experienced sudden, massive spikes in Google Trends searches within Indonesia and Southeast Asia. These spikes did not always correspond to major real-time physical events or conventional coverage by mainstream national news outlets. Instead, they were largely driven by automated recommendation algorithms on search engines, news apps, and social media platforms. These machine learning models analyze real-time engagement and search patterns, rapidly amplifying specific keywords and distributing them to millions of feeds in a self-reinforcing loop. This dynamic illustrates a broader shift in media consumption, where public attention is increasingly shaped by predictive recommendation algorithms rather than traditional editorial curation. This high-velocity digital culture is reflected in the trending media consumed by the Indonesian public. A detailed snapshot of YouTube trends on June 22, 2026, reveals a mix of highly localized television dramas, digital gaming, regional sports, and viral music.

Trend Category	Dominant Content / Titles	Key Cultural Drivers
Broadcast Drama	<i>Terikat Janji</i> (Episodes 79 and associated promotional trailers starring Arya Saloka and Asha Assuncao).	High viewer engagement with domestic television narratives, amplified by social media commentary and fan communities.
Digital Gaming	<i>Roblox</i> and <i>Minecraft</i> gameplay videos (e.g., "Mipan & Zuzuzu" and "Yuta Mio Panik" series).	Large youth demographics driving high view counts for cooperative, sandbox, and roleplay gaming content.
Regional Sports	<i>2026 Asian Games Esports Qualifiers</i> (specifically Mobile Legends: Bang Bang and PUBG Mobile finals).	Growing institutional support for competitive esports across Southeast Asia.
Domestic Music	Live music videos and acoustic covers featuring regional artists like Ajeng Febria and Tri Suaka.	Strong public interest in localized pop and traditional folk-pop fusions.

These patterns of digital consumption coexisted with major national economic and environmental developments in mid-2026. On June 17, 2026, the Indonesian Rupiah strengthened by 0.76% from its May levels, closing at Rp17,730 per U.S. Dollar. Bank Indonesia Governor Perry Warjiyo attributed this stabilization to rising foreign portfolio investment in Bank Indonesia Rupiah Securities (SRBI). Outstanding SRBI holdings reached Rp1,021.1 trillion, with non-resident ownership rising to Rp238.1 trillion (23.3% of the total), drawing foreign capital through attractive yields on 6, 9, and 12-month tenors.

At the same time, the Ministry of Manpower's *Employment Outlook 2026* projected that the transition to a green economy would create approximately 3.88 million jobs across Indonesia. This employment growth is supported by efforts to expand natural resource processing, develop renewable energy, and advance transportation electrification.

However, this economic transition occurred alongside ongoing environmental challenges. In the 24 hours leading up to June 23, 2026, the Ministry of Environment and Forestry (KLHK) satellite

network detected 520 active wildfire hotspots across the archipelago, with the highest concentrations located in Maluku, East Kalimantan, and North Maluku. Additionally, on the local level, municipal administrators worked to balance development with public spaces. During the June 2026 school holidays, the city of Bandung promoted several free public parks—such as Taman Balai Kota, Taman Sejarah, and the Babakan Siliwangi forest walk—to provide accessible recreation options and support urban biodiversity.

Comparative Structural Synthesis

Managing modern socio-technical systems requires balancing individual user experiences with broader systemic governance. At the individual level, conversational design must focus on clarity, context, and simplicity to ensure that automated interfaces remain intuitive and accessible. At the systemic level, digital and resource platforms require robust verification layers, such as the Nexus Truth Engine, to prevent algorithmic erasures, misinformation, or policy failures from compounding.

Whether addressing biological mutations, historical records, or regional economic transitions, the design of these platforms determines what information is preserved, who is protected, and how resources are managed. Only by integrating user-centered design with rigorous governance frameworks can organizations ensure these automated systems remain reliable, accountable, and aligned with human needs.

Works cited

1. What is Conversational AI? | Globant Tech Terms, <https://www.globant.com/tech-terms/conversational-ai>
2. What is Conversational AI? | A Practical Guide - K2view, <https://www.k2view.com/what-is-conversational-ai/>
3. Conversational Assistant - CubeworkFreight & Logistics Glossary | item.com, <https://www.item.com/pt/glossary/conversational-assistant>
4. Home Assistant Voice PE (AI Agent): Start Conversations with conversation.process and conversation_id, <https://community.home-assistant.io/t/home-assistant-voice-pe-ai-agent-start-conversations-with-conversation-process-and-conversation-id/862489>
5. A Survey on Multi-User Conversational Interfaces - MDPI, <https://www.mdpi.com/2076-3417/15/13/7267>
6. Intents | Conversational Actions - Google for Developers, <https://developers.google.com/assistant/conversational/intents>
7. Confirmations - Conversation Design | Google for Developers, <https://developers.google.com/assistant/conversation-design/confirmations>
8. 11 Conversational Design - Digital Assistant - Oracle Help Center, <https://docs.oracle.com/en/cloud/paas/digital-assistant/use-chatbot/conversational-design.html>
9. Conversational Design - UX Magazine, <https://uxmag.com/articles/conversational-design-banner>
10. Talking AI: how conversational styles shape consumer attitudes in hedonic and utilitarian contexts | Marketing Intelligence & Planning | Emerald Publishing, <https://www.emerald.com/mip/article/doi/10.1108/MIP-03-2025-0206/1361196/Talking-AI-how-conversational-styles-shape>
11. Algorithmic Management in a Scandinavian context - CBS Research Portal, https://research.cbs.dk/files/119276026/2098542_167822_Algorithmic_Management_in_a_Scandinavian_Context.pdf
12. The Experience of Adoption - YouTube Music, <https://music.youtube.com/playlist?list=PLke4yM5Y3qc8-1u0zvw7MhMgKZos85t0f>
13. Efficient

Non-Viral Ablation of COL7A1 Exon 73 Splice Acceptor for the Treatment of Dystrophic Epidermolysis Bullosa (DEB) - Factor Bioscience, <https://factor.bio/wp-content/uploads/2020/02/Efficient-Non-Viral-Ablation-of-COL7A1-Exon-73-Splice-Acceptor-for-the-Treatment-of-Dystrophic-Epidermolysis-Bullosa-DEB.pdf> 14. Unsettling Memories: Practice, Platforms, and Possibilities. Closing Reflections on the Communicating Memory Matters Collection | Memory, Mind & Media | Cambridge Core, <https://www.cambridge.org/core/journals/memory-mind-and-media/article/unsettling-memories-practice-platforms-and-possibilities-closing-reflections-on-the-communicating-memory-matters-collection/D9C4B6836FF4A410C60BDA6AFF44C290> 15. (PDF) Unsettling Memories: Practice, Platforms, and Possibilities. Closing Reflections on the Communicating Memory Matters Collection - ResearchGate, https://www.researchgate.net/publication/398697633_Unsettling_Memories_Practice_Platforms_and_Possibilities_Closing_Reflections_on_the_Communicating_Memory_Matters_Collection 16. The Experience of Adoption - Émission - Apple Podcasts, <https://podcasts.apple.com/us/podcast/the-experience-of-adoption/id1888390389?l=fr-FR> 17. Shane Bouel, <https://bouel.com.au/wp-content/uploads/2021/08/Resume-2021-2.pdf> 18. Shane Bouel | CREATIVE MENTOR + DESIGNER | Welcome, <https://bouel.com.au/> 19. 2026: The Year the System Stops Pretending | by Shane Bouel | Thoughtless Delineation, <https://medium.com/thoughtless-delineation/2026-the-year-the-system-stops-pretending-193fc41f5edd> 20. The Experience of Adoption - Podcast - Apple Podcasts, <https://podcasts.apple.com/us/podcast/the-experience-of-adoption/id1888390389?l=es-MX> 21. Tanggung Jawab Perdata Penyedia Platform Media Sosial atas Kerugian Pengguna Akibat Penyebaran Konten Deepfake yang Tidak Dimoderasi - Scribd, <https://id.scribd.com/document/1050594033/Tanggung-Jawab-Perdata-Penyedia-Platform-Media-Sosial-atas-Kerugian-Pengguna-Akibat-Penyebaran-Konten-Deepfake-yang-Tidak-Dimoderasi> 22. Decoding the Wickedness of Resource Nexus Problems—Examples from Water-Soil Nexus Problems in China - MDPI, <https://www.mdpi.com/2079-9276/7/4/67> 23. The Urban Water–Energy Nexus: Building Resilience for Global Change in the “Urban Century” | Request PDF - ResearchGate, https://www.researchgate.net/publication/300076813_The_Urban_Water-Energy_Nexus_Building_Resilience_for_Global_Change_in_the_Urban_Century 24. (PDF) Urban Solid Waste Management: Case of Ghana in West Africa - ResearchGate, https://www.researchgate.net/publication/283045438_Urban_Solid_Waste_Management_Case_of_Ghana_in_West_Africa 25. UNESCO World Heritage and the SDGs – Interdisciplinary Perspectives - Academia.edu, https://www.academia.edu/91495817/UNESCO_World_Heritage_and_the_SDGs_Interdisciplinary_Perspectives 26. XIV. Nexus Truth Engine | Organization, <https://docs.therisk.global/organization/organization/architecture/ii.-definitions/xiv.-nexus-truth-engine> 27. David Alaba Trending Dunia: Bagaimana AI & Data Intelligence Mengubah Narasi Atlet Global di 2026, <https://bernas.id/nasional/news/david-alaba-trending-dunia-bagaimana-ai-data-intelligence-mengubah-narasi-atlet-2026-06-22> 28. Foreign inflows into SRBI help stabilize rupiah: BI Governor, <https://en.antaranews.com/news/419663/foreign-inflows-into-srbi-help-stabilize-rupiah-bi-governor> 29. Ministry projects 3.88 million green jobs in Indonesia in 2026, <https://en.antaranews.com/news/419999/ministry-projects-388-million-green-jobs-in-indonesia-in-2026> 30. Edin Dzeko Trending Indonesia: Apa Konteks Berita yang Sebenarnya? - BERNAS.id, <https://bernas.id/papua/news/edin-dzeko-trending-indonesia-apa-konteks-berita-yang-sebenarn>

ya-2026-06-12 31. Music Videos Trending in Indonesia - Kworb.net,
<https://kworb.net/youtube/trending/id.html> 32. 520 Hotspots Detected in Indonesia Over Last 24
Hours (Tuesday, 23 June 2026),
<https://databoks.katadata.co.id/en/demographics/statistics/3e2efa494e4b8bb/520-hotspots-detected-in-indonesia-over-last-24-hours-tuesday-23-june-2026> 33. Short Escape Hemat! Ini 5
Taman Estetik dan Gratis di Bandung Buat Isi Liburan Sekolah Juni 2026,
<https://style.tribunnews.com/life/373208/short-escape-hemat-ini-5-taman-estetik-dan-gratis-di-bandung-buat-isi-liburan-sekolah-juni-2026>